

Exploratory Data Analysis Report

Dataset Overview

The dataset contains multiple numerical and categorical variables. Some columns contain missing values. Data types include integers, floats, and object variables.

Statistical Findings

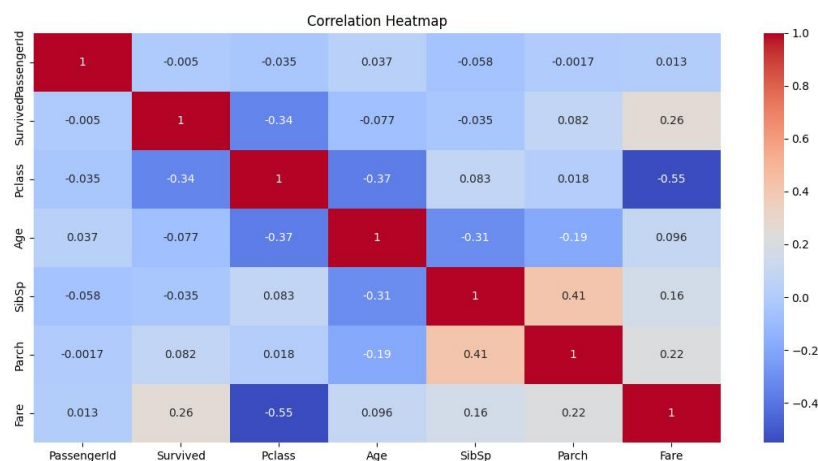
Numerical variables exhibit varying ranges and distributions. Some features contain significant outliers. Certain variables are positively skewed.

Missing Values

Missing values exist in several columns. Data cleaning is required before model building.

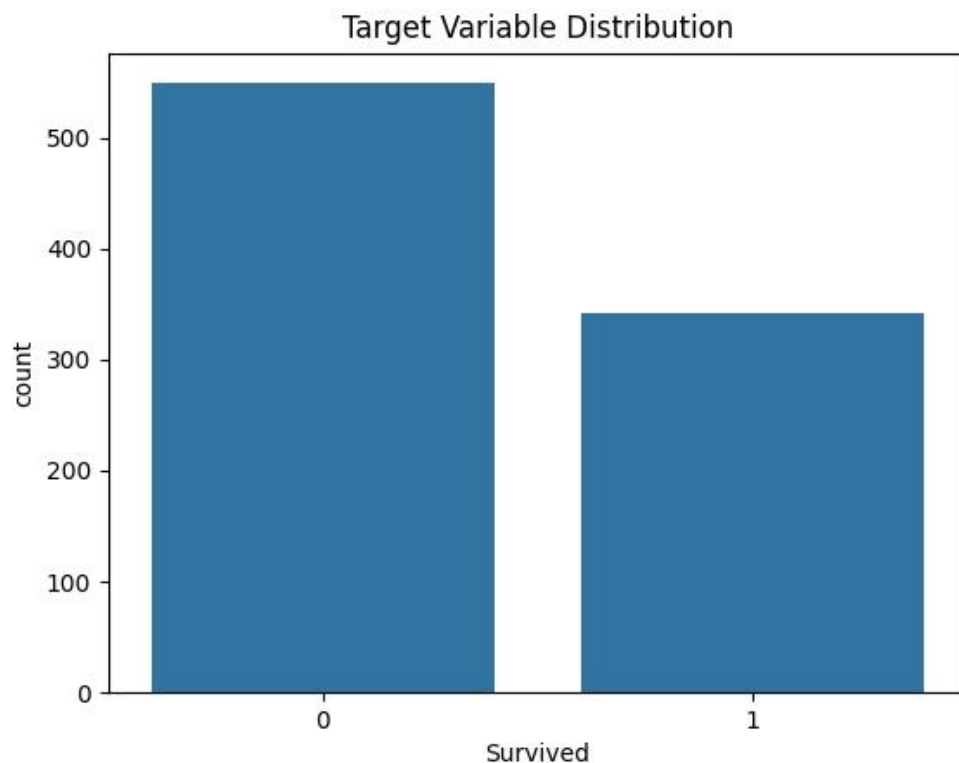
Correlation Analysis

Heatmap shows relationships among numerical variables. Some variables are highly correlated. Highly correlated features may introduce multicollinearity.



Distribution Analysis

Histograms indicate that several variables are not normally distributed. KDE plots reveal skewness in some features.



Outlier Analysis

Boxplots identify outliers in multiple numerical variables. Outliers may affect prediction accuracy.

Relationship Analysis

Pairplot and scatter plots reveal trends between numerical variables. Some variables show positive correlation while others have weak relationships.

Categorical Analysis

Count plots show frequency distribution of categorical variables. Certain categories dominate the dataset, indicating class imbalance.

Overall Findings

The dataset contains missing values and outliers that require preprocessing. Numerical variables have different distributions and scales. Correlation analysis suggests some strongly related features. Visualizations help identify trends and potential predictive variables. Data preprocessing such as missing value treatment, encoding categorical variables, scaling, and outlier handling is recommended before machine learning.